Chapter 8: An Introduction to Metabolism

Complete Master Reference Edition • Fused Curriculum (Introduction, Metabolism, Energy, Enzymes, Lecture Slides & Practice Questions)

Course: BSC2010C General Biology I

Institution: University of Central Florida

Source: Canvas Modules, Instructor Slides & Active Learning #3

Learning Objectives

- Understand different forms of energy
- Understand the First and Second Law of Thermodynamics
- Differentiate between endergonic and exergonic reactions
- Predict energy coupling between endergonic and exergonic reactions
- Describe a catalyst
- Explain how enzymes are catalysts
- Describe the enzyme substrate complex
- Distinguish between different type of enzymatic inhibitors

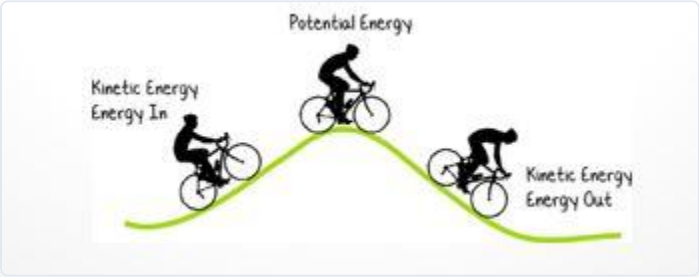
Chapter 8: An Introduction to Metabolism

We are accustomed to hearing the word "metabolism" as it relates to our bodies; I am sure you know of someone who eats large quantities of food but never gains a pound, or someone who lives on salad but is overweight. How your body processes food, how much energy your body acquires from that food, and how much energy your body needs to maintain your health is unique to you; it is your metabolism.

Individual cells, whether a single-celled organism or one cell within a complex organ such as the liver, also have a metabolism. For a cell, it's metabolism is all the chemical processes it has to carry out to maintain life. Food has to be digested, broken down from large polymers of carbohydrates or proteins into small monomers of glucose or amino acids. Those monomers are then reassembled into different proteins or carbohydrates the cell will use. Cellular respiration is the process by which cells obtain energy from sugars and other forms of fuel. Cells can then use that energy to build larger polymers, to fuel chemical reactions, to actively transport molecules across the plasma membrane, etc.

Before we learn more about how cells obtain and use energy, we need to learn about energy. Specifically, you are going to learn about three different energy forms: Kinetic Energy, Potential Energy, and Chemical Energy (which is a type of Potential Energy), then you will learn about the first two Laws of Thermodynamics.

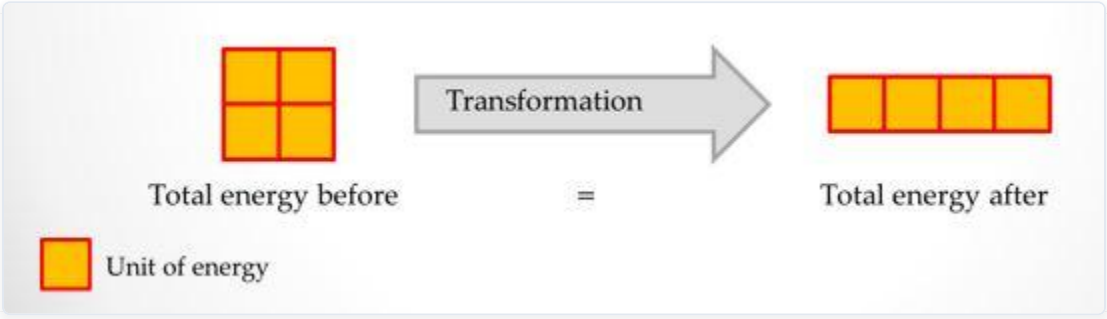
Kinetic Energy is the energy associated with motion. Think about riding a bike uphill. You have to put energy into pedaling to get to the top of the hill - this is putting kinetic energy in (you had to work to get to the top of the hill). Let's say you pedaled so hard to get to the top of the hill that you stopped to catch your breath before coasting down the hill. **Potential Energy** is energy that is stored due to location or position. At the top of the hill, you and the bike have stored all the energy you put into getting there - that energy is stored as potential energy. As you coast down the hill you are converting that potential energy into kinetic energy - this is kinetic energy out (think about how fast you are going at the bottom of the hill). Looking at this entire "reaction": You put kinetic energy in to get to the top of the hill, your maximum potential energy is at the top of the hill, you get kinetic energy out by coasting down the hill without having to pedal.



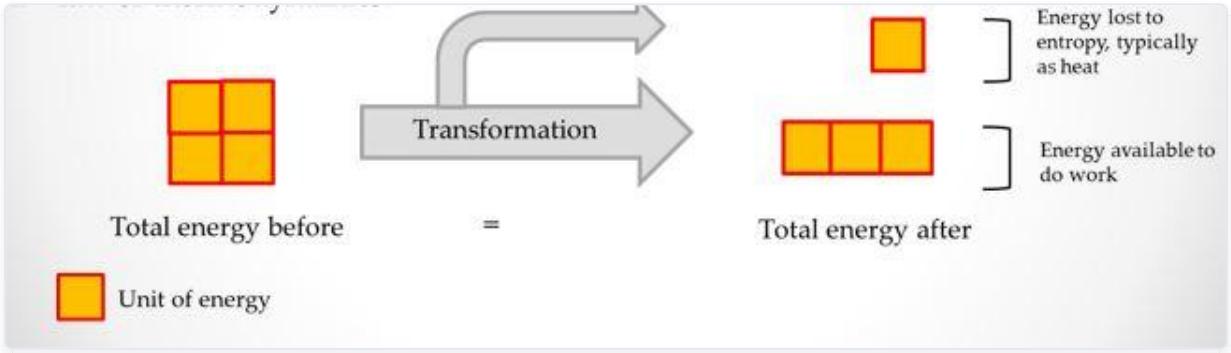
Chemical Energy is the potential energy stored in chemical bonds. Breaking certain types of chemical bonds will release that energy. Cells get energy by hydrolyzing the bonds of carbohydrates, proteins, and lipids. The gasoline you put in your car is also a type of chemical energy. You car breaks the bonds in the gasoline to get the energy that moves your car.

Energy practice

The **First Law of Thermodynamics** states that energy can be transferred or transformed but cannot be created or destroyed. In other words, the total amount of energy in the universe is constant. We can use the same gasoline example from above: gasoline is chemical energy that your car converts to kinetic energy that causes the car to move. The gas company did not create the energy in the gasoline and your car does not destroy the energy in gasoline - it was simply converted from one form to another.



The Second Law of Thermodynamics states that every energy transfer or transformation increases the entropy (disorder) of the universe; the transfer or transformation is not 100% efficient. Continuing with the car/gasoline example, this means that when your car converts the chemical energy of gasoline into kinetic energy of motion, some energy is lost to heat. Your engine gets hot, right? That heat is the result of converting gasoline into motion. In organisms, some energy is also lost to heat.



Laws of Thermodynamics practice

CHAPTER READING

Chapter 8 textbook reading: Metabolism, Energy, and Enzymes

CHAPTER LINKS

Chapter 8 PowerPoint

Chapter 8 Post-Quiz

Chapter 8 Simple Practice Problems & Extra Reaction Coupling Problems (and the answers)

Section 8.1: Cellular Metabolism & Metabolic Pathways

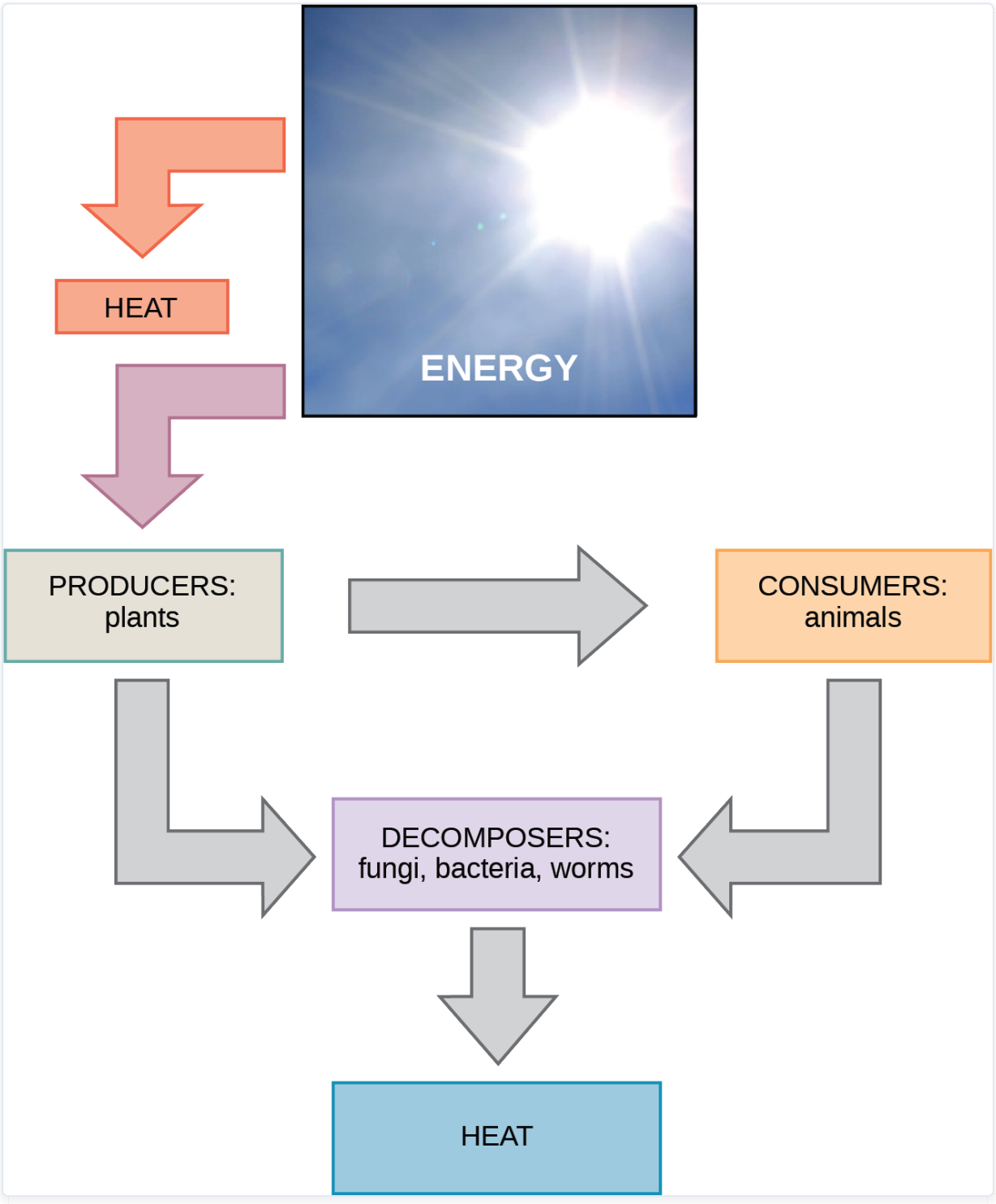
Bioenergetics, carbohydrate metabolic cycles, anabolic vs. catabolic pathway dynamics, and ATP currency.

Learning Objectives

By the end of this section, you will be able to do the following:

- Explain metabolic pathways and describe the two major types
- Discuss how chemical reactions play a role in energy transfer

Scientists use the term bioenergetics to discuss the concept of energy flow (Figure 6.2) through living systems, such as cells. Cellular processes such as building and breaking down complex molecules occur through stepwise chemical reactions. Some of these chemical reactions are spontaneous and release energy; whereas, others require energy to proceed. Just as living things must continually consume food to replenish what they have used, cells must continually obtain more energy to replenish that which the many energy-requiring chemical reactions that constantly take place use. All of the chemical reactions that transpire inside cells, including those that use and release energy, are the cell's metabolism.



This diagram shows energy from the sun being transferred to producers, such as plants, as well as releasing heat. The producers in turn transfer the energy to consumers and decomposers, which release heat. Animals also t...

Figure 6.2 Most life forms on earth obtain their energy from the sun. Plants use photosynthesis to capture sunlight, and herbivores eat those plants to obtain energy. Carnivores eat the herbivores, and decomposers digest plant and animal matter.

Carbohydrate Metabolism

Sugar metabolism is a classic example of the many cellular processes that use and produce energy. Living things consume sugar as a major energy source, because sugar molecules have considerable energy stored within their bonds. The following equation describes the breakdown of glucose, a simple sugar:

$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + \text{energy}$

Consumed carbohydrates have their origins in photosynthesizing organisms like plants (Figure 6.3). During photosynthesis, plants use the energy of sunlight to convert carbon dioxide gas (CO_2) into sugar molecules, like glucose ($C_6H_{12}O_6$). Because this process involves synthesizing a larger, energy-storing molecule, it requires an energy input to proceed. The following equation (notice that it is the reverse of the previous equation) describes the synthesis of glucose:

$6CO_2 + 6H_2O + \text{energy} \rightarrow C_6H_{12}O_6 + 6O_2$

During photosynthesis chemical reactions, energy is in the form of a very high-energy molecule scientists call ATP, or adenosine triphosphate. This is the primary energy currency of all cells. Just as the dollar is the currency we use to buy goods, cells use ATP molecules as energy currency to perform immediate work. The sugar (glucose) is stored as starch or glycogen. Energy-storing polymers like these break down into glucose to supply ATP molecules.

Metabolic Pathways

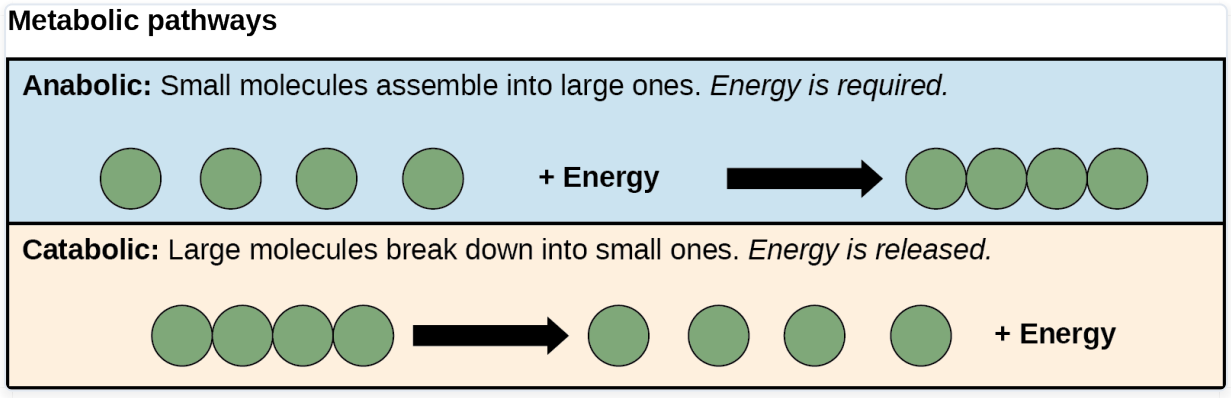
The processes of making and breaking down sugar molecules illustrate two types of metabolic pathways. A metabolic pathway is a series of interconnected biochemical reactions that convert a substrate molecule or molecules, step-by-step, through a series of metabolic intermediates, eventually yielding a final product or products. In the case of sugar metabolism, the first metabolic pathway synthesized sugar from smaller molecules, and the other pathway broke sugar down into smaller molecules. Scientists call these two opposite processes—the first requiring energy and the second producing energy—anabolic (building) and catabolic (breaking down) pathways.

Anabolic and Catabolic Pathways

Anabolic pathways require an input of energy to synthesize complex molecules from simpler ones. Synthesizing large proteins from amino acid building blocks, and synthesizing new DNA strands from nucleic acid building blocks are examples of anabolic pathways. These biosynthetic processes are critical to the cell's life, take place constantly, and demand energy that ATP provides.

ATP is an important molecule for cells to have in sufficient supply at all times. The breakdown of sugars illustrates how a single glucose molecule can store enough energy to make a great deal of ATP. This is a catabolic pathway. Catabolic pathways involve degrading (or breaking down) complex molecules into simpler ones. Molecular energy stored in complex molecule bonds release in catabolic pathways and harvest in such a way that it can produce ATP. Other energy-storing molecules, such as fats, also break down through similar catabolic reactions to release energy and make ATP (Figure 6.5).

It is important to know that metabolic pathway chemical reactions do not take place spontaneously. A protein called an enzyme facilitates or catalyzes each reaction step. Enzymes are important for catalyzing all types of biological reactions—those that require energy as well as those that release energy.



Anabolic and catabolic pathways are shown. In the anabolic pathway (top), four small molecules have energy added to them to make one large molecule. In the catabolic pathway (bottom), one large molecule is broken down in...

Figure 6.5 Anabolic pathways are those that require energy to synthesize larger molecules. Catabolic pathways are those that generate energy by breaking down larger molecules. Both types of pathways are required for maintaining the cell's energy balance.

Section 8.2: Energy Types, Thermodynamics & Free Energy (Delta G)

Kinetic vs. potential energy, Gibbs free energy calculations, exergonic/endergonic thermodynamics, and activation energy.

Learning Objectives

By the end of this section, you will be able to do the following:

- Define “energy”
- Explain the difference between kinetic and potential energy
- Discuss the concepts of free energy and activation energy
- Describe endergonic and exergonic reactions

We define energy as the ability to do work. Energy exists in different forms. For example, electrical energy, light energy, and heat energy are all different energy types. While these are all familiar energy types that one can see or feel, there is another energy type that is much less tangible, potential energy. Scientists associate this energy with something as simple as an object above the ground. In order to appreciate the way energy flows into and out of biological systems, it is important to understand more about the different energy types that exist in the physical world.

Energy Types

When an object is in motion, there is energy. For example, an airplane in flight produces considerable energy. This is because moving objects are capable of enacting a change, or doing work. Think of a wrecking ball. Even a slow-moving wrecking ball can do considerable damage to other objects. However, a wrecking ball that is not in motion is incapable of performing work. Energy associated with objects in motion is **kinetic energy**. A speeding bullet, a walking person, rapid molecule movement in the air (which produces heat), and electromagnetic radiation like light all have kinetic energy.

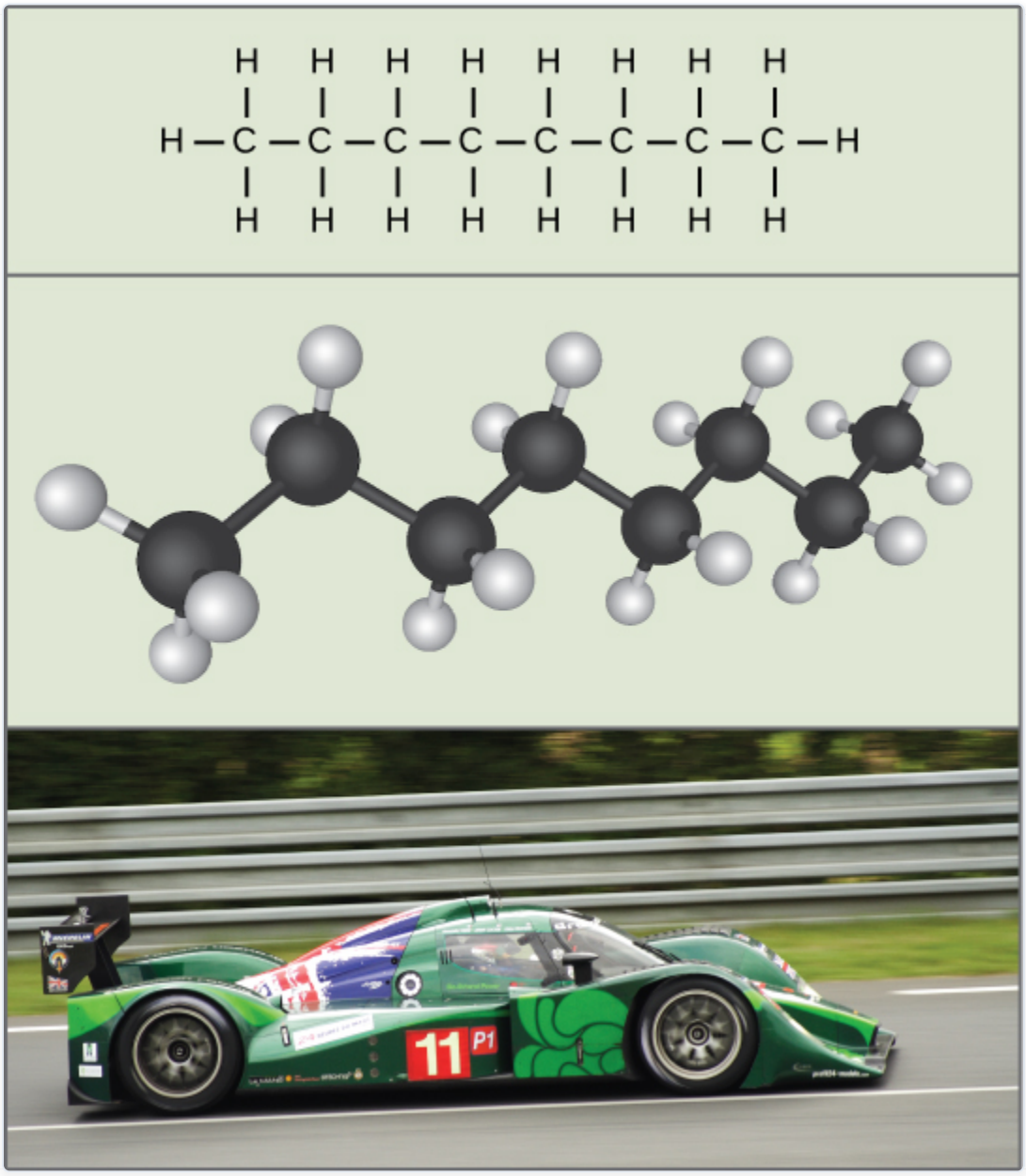
What if we lift that same motionless wrecking ball two stories above a car with a crane? If the suspended wrecking ball is unmoving, can we associate energy with it? The answer is yes. The suspended wrecking ball has associated energy that is fundamentally different from the kinetic energy of objects in motion. This energy form results from the ***potential*** for the wrecking ball to do work. If we release the ball it would do work. Because this energy type refers to the potential to do work, we call it **potential energy**. Objects transfer their energy between kinetic and potential in the following way: As the wrecking ball hangs motionless, it has 0 kinetic and 100 percent potential energy. Once it releases, its kinetic energy begins to increase because it builds speed due to gravity. Simultaneously, as it nears the ground, it loses potential energy. Somewhere mid-fall it has 50 percent kinetic and 50 percent potential energy. Just before it hits the ground, the ball has nearly lost its potential energy and has near-maximal kinetic energy. Other examples of potential energy include water's energy held behind a dam (Figure 6.6), or a person about to skydive from an airplane.



The photo on the left shows a river that is blocked by a giant cement wall, called a dam. The photo on the right shows a waterfall.

Figure 6.6 Water behind a dam has potential energy. Moving water, such as in a waterfall or a rapidly flowing river, has kinetic energy. (credit "dam": modification of work by "Pascal"/Flickr; credit "waterfall": modification of work by Frank Gualtieri)

We associate potential energy not only with the matter's location (such as a child sitting on a tree branch), but also with the matter's structure. A spring on the ground has potential energy if it is compressed; so does a tautly pulled rubber band. The very existence of living cells relies heavily on structural potential energy. On a chemical level, the bonds that hold the molecules' atoms together have potential energy. Remember that anabolic cellular pathways require energy to synthesize complex molecules from simpler ones, and catabolic pathways release energy when complex molecules break down. That certain chemical bonds' breakdown can release energy implies that those bonds have potential energy. In fact, there is potential energy stored within the bonds of all the food molecules we eat, which we eventually harness for use. This is because these bonds can release energy when broken. Scientists call the potential energy type that exists within chemical bonds that releases when those bonds break chemical energy (Figure 6.7). Chemical energy is responsible for providing living cells with energy from food. Breaking the molecular bonds within fuel molecules brings about the energy's release.



The molecular formula of octane (top), which is a chain of eight carbons and eighteen hydrogens, fuels a racecar speeding along a track (bottom).

Figure 6.7 The molecules in gasoline contain chemical energy within the chemical bonds. This energy transforms into kinetic energy that allows a car to race on a racetrack. (credit "car": modification of work by Russell Trow)

Free Energy

After learning that chemical reactions release energy when energy-storing bonds break, an important next question is how do we quantify and express the chemical reactions with the associated energy? How can we compare the energy that releases from one reaction to that of another reaction? We use a measurement of free energy to quantitate these energy transfers. Scientists call this free energy Gibbs free energy (abbreviated with the letter G) after Josiah Willard Gibbs, the scientist who developed the measurement. According to the second law of thermodynamics, all energy transfers involve losing some energy in an unusable form such as heat, resulting in entropy. Gibbs free energy specifically refers to the energy that takes place with a chemical reaction that is available

after we account for entropy. In other words, Gibbs free energy is usable energy, or energy that is available to do work.

Every chemical reaction involves a change in free energy, called delta G (ΔG). We can calculate the change in free energy for any system that undergoes such a change, such as a chemical reaction.

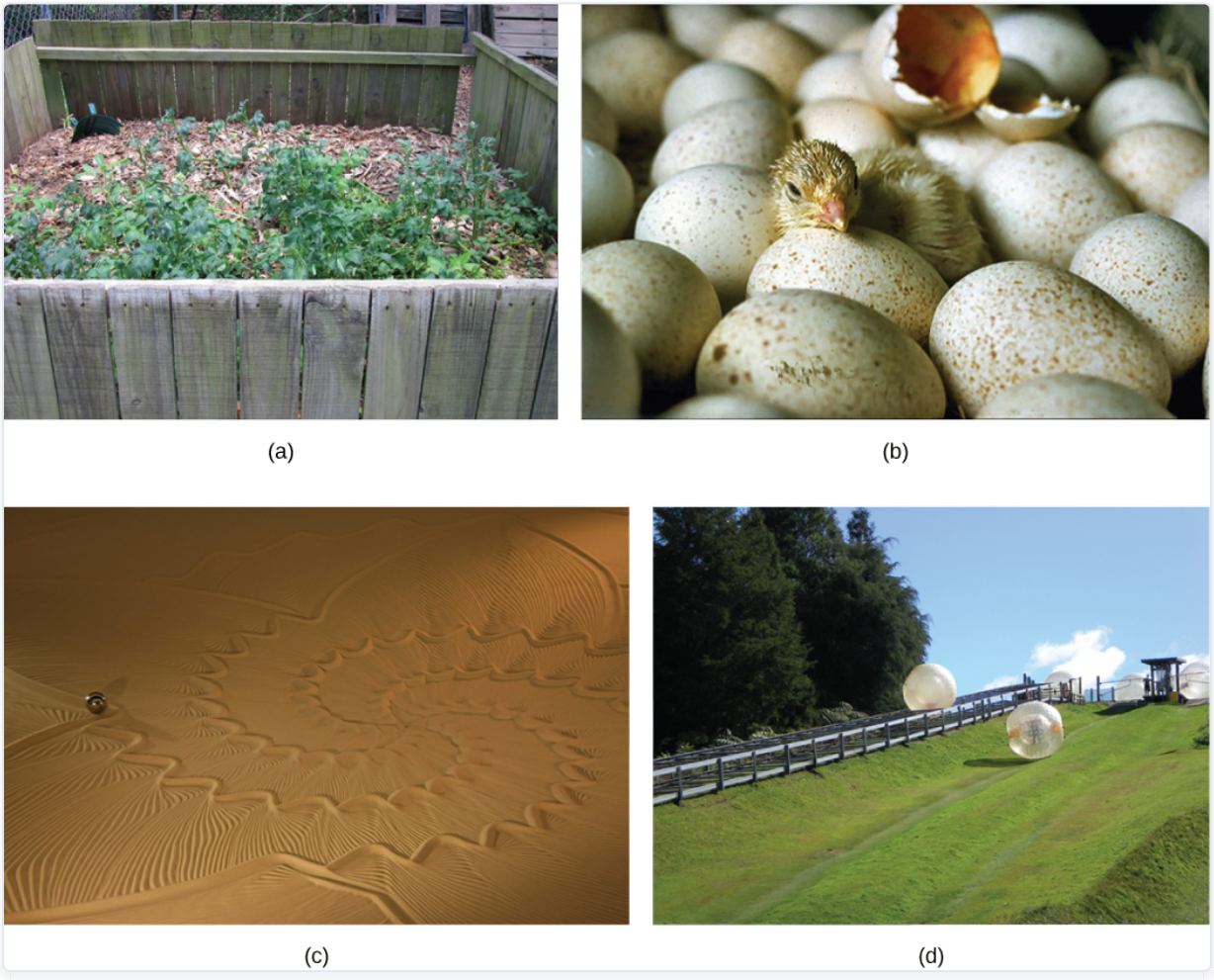
Endergonic Reactions and Exergonic Reactions

If energy releases during a chemical reaction, then the resulting value from the above equation will be a negative number. In other words, reactions that release energy have a **ΔG < 0**. A negative ΔG also means that the reaction's products have less free energy than the reactants, because they gave off some free energy during the reaction. Scientists call reactions that have a negative ΔG and consequently release free energy **exergonic reactions.** Think: *ex*ergonic means energy is *ex*iting the system. We also refer to these reactions as spontaneous reactions, because they can occur without adding energy into the system. Understanding which chemical reactions are spontaneous and release free energy is extremely useful for biologists, because these reactions can be harnessed to perform work inside the cell. We must draw an important distinction between the term spontaneous and the idea of a chemical reaction that occurs immediately. Contrary to the everyday use of the term, a spontaneous reaction is not one that suddenly or quickly occurs. Rusting iron is an example of a spontaneous reaction that occurs slowly, little by little, over time.

If a chemical reaction requires an energy input rather than releasing energy, then the **ΔG for that reaction will be a positive** value. In this case, the products have more free energy than the reactants. Thus, we can think of the reactions' products as energy-storing molecules. We call these chemical reactions endergonic reactions, and they are non-spontaneous. An endergonic reaction will not take place on its own without adding free energy.

Let's revisit the example of the synthesis and breakdown of the food molecule, glucose. Remember that building complex molecules, such as sugars, from simpler ones is an anabolic process and requires energy. Therefore, the chemical reactions involved in anabolic processes are endergonic reactions. Alternatively the catabolic process of breaking sugar down into simpler molecules releases energy in a series of exergonic reactions. Like the rust example above, the sugar breakdown involves spontaneous reactions, but these reactions do not occur instantaneously. Figure 6.8 shows some other examples of endergonic and exergonic reactions.

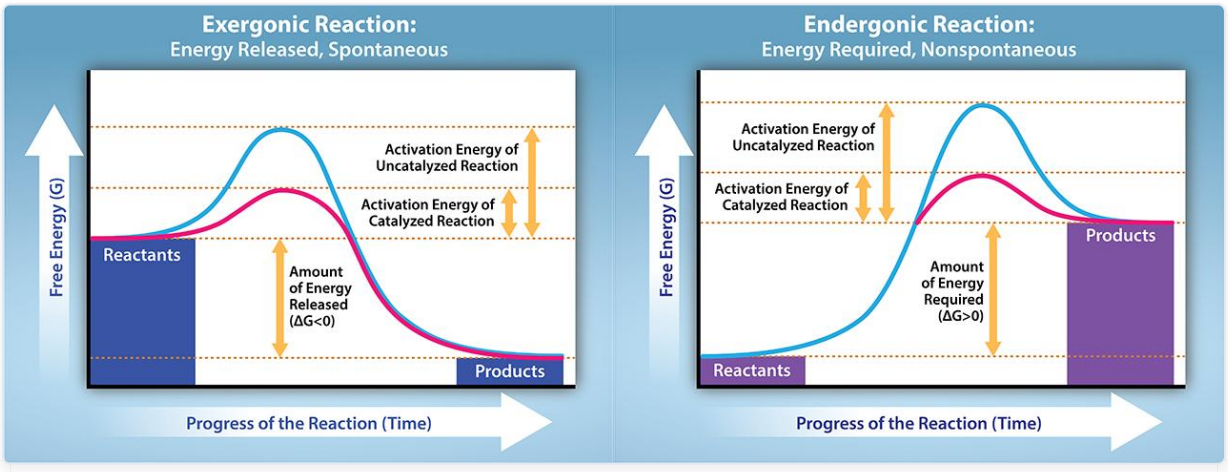
VISUAL CONNECTION



There are four photos show. The first photo shows a pile of wood chips and dirt, with small plants growing from this. The second photo shows a small baby bird breaking out of its egg as it hatches. The third photo sho...

Figure 6.8 This figure shows some examples of endergonic processes (ones that require energy) and exergonic processes (ones that release energy). These include (a) a compost pile decomposing, (b) a chick developing from a fertilized egg, (c) sand art destruction, and (d) a ball rolling down a hill. (credit a: modification of work by Natalie Maynor; credit b: modification of work by USDA; credit c: modification of work by "Athlex"/Flickr; credit d: modification of work by Harry Malsch)

Look at each of the images above and decide if the processes shown are endergonic or exergonic.



The two plots show the change in Gibbs free energy as reactants are converted to products. Gibbs free energy decreases with time for an exergonic reaction (left), and the reaction is spontaneous. Gibbs free energy increa...

Figure 6.9 Exergonic and endergonic reactions result in changes in Gibbs free energy (G). Exergonic reactions have a net release of energy and are spontaneous reactions. Endergonic reactions require an input of energy to proceed and are nonspontaneous reactions. Both Exergonic and endergonic reactions require initial activation energy for the reaction to occur.Credit: Tag, A., Rao, A., Fletcher, S. and Ryan, A. Department of Biology, Texas A&M University.

Activation Energy

There is another important concept that we must consider regarding endergonic and exergonic reactions. Even exergonic reactions require a small amount of energy input before they can proceed with their energy-releasing steps. These reactions have a net release of energy, but still require some initial energy. Scientists call this small amount of energy input necessary for all chemical reactions to occur the activation energy (or free energy of activation) abbreviated as E_A (Figure 6.10).

Why would an energy-releasing, negative ΔG reaction actually require some energy to proceed? The reason lies in the steps that take place during a chemical reaction. During chemical reactions, certain chemical bonds break and new ones form. For example, when a glucose molecule breaks down, bonds between the molecule's carbon atoms break. Since these are energy-storing bonds, they release energy when broken. However, to get them into a state that allows the bonds to break, the molecule must be somewhat contorted. A small energy input is required to achieve this contorted state. This contorted state is the transition state, and it is a high-energy, unstable state. For this reason, reactant molecules do not last long in their transition state, but very quickly proceed to the chemical reaction's next steps.

Section 8.3: Enzymes as Molecular Catalysts & Allosteric Regulation

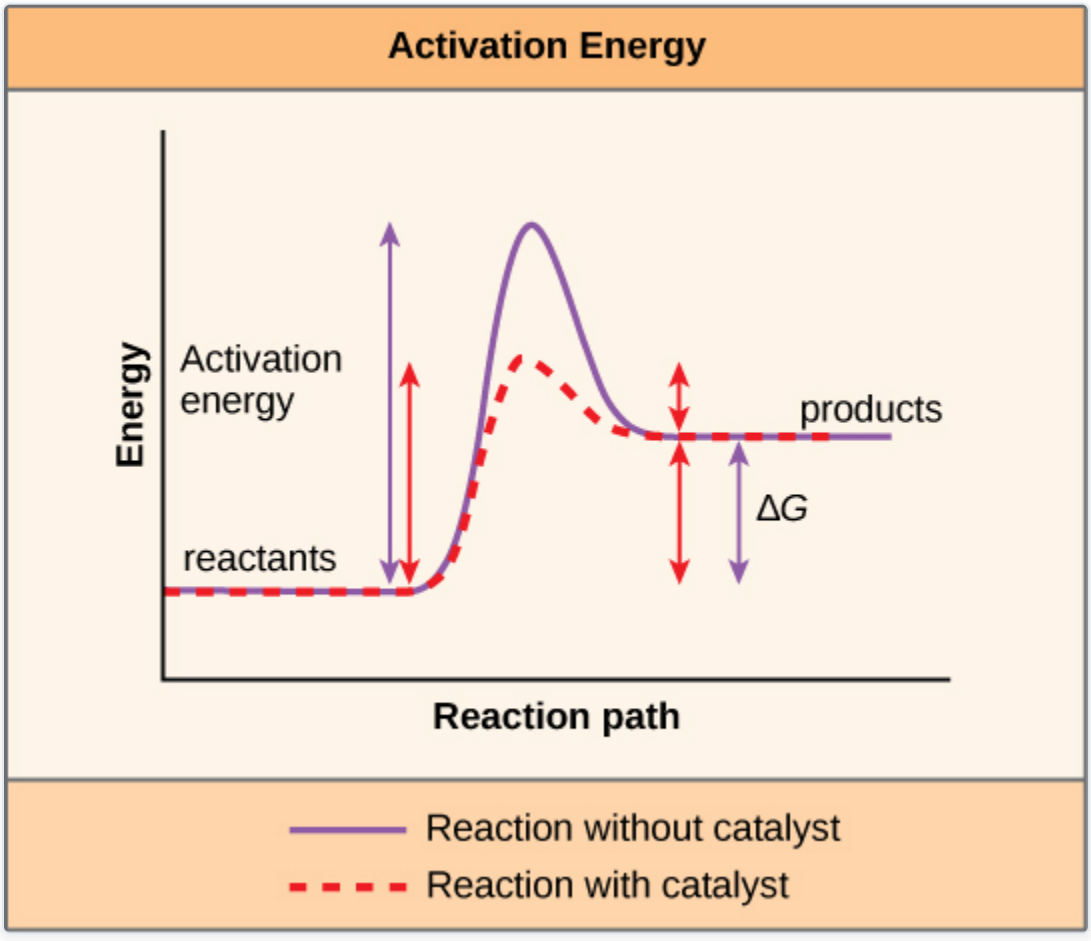
Active site specificity, induced-fit dynamics, temperature/pH denaturation, competitive/noncompetitive inhibition, and feedback control.

Learning Objectives

By the end of this section, you will be able to do the following:

- Describe the role of enzymes in metabolic pathways
- Explain how enzymes function as molecular catalysts
- Discuss enzyme regulation by various factors

A substance that helps a chemical reaction to occur is a catalyst, and the special molecules that catalyze biochemical reactions are **enzymes.** All enzymes are proteins, comprised of amino acid chains, and they perform the critical task of lowering the activation energies of chemical reactions inside the cell. Enzymes do this by binding to the reactant molecules, and holding them in such a way as to make the chemical bond-breaking and bond-forming processes take place more readily. It is important to remember that enzymes do not change the reaction's ΔG. In other words, they do not change whether a reaction is exergonic or endergonic. They only reduce the activation energy required to reach the transition state (Figure 6.15).



This plot shows that a catalyst decreases the activation energy for a reaction but does not change the Gibbs free energy.

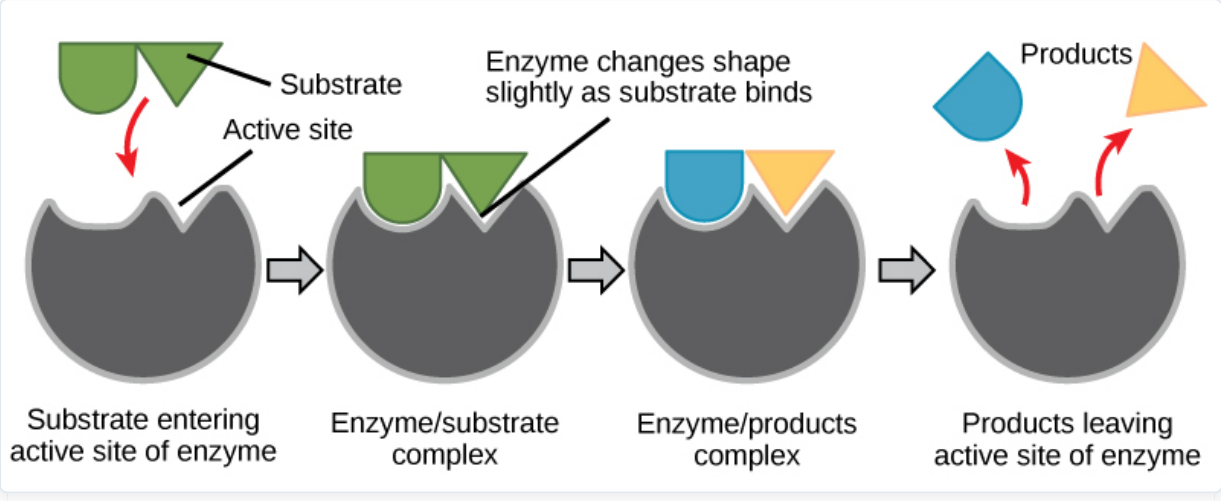
Figure 6.15 Enzymes lower the reaction's activation energy but do not change the reaction's free energy.

Enzyme Active Site and Substrate Specificity

The chemical reactants to which an enzyme binds are the enzyme's **substrates**. There may be one or more substrates, depending on the particular chemical reaction. In some reactions, a single-reactant substrate breaks down into multiple products. In others, two substrates may come together to create one larger molecule. The location within the enzyme where the substrate binds is the enzyme's **active site**. This is where the "action" happens. Since enzymes are proteins, there is a unique combination of amino acids, their hydrogen bonds and R group interactions within the active site. These can be large or small, weakly acidic or basic, hydrophilic or hydrophobic, positively or negatively charged, or neutral. The unique combination of amino acids, their positions, sequences, structures, and properties, creates a very specific chemical environment within the active site. This specific environment is suited to bind to a specific chemical substrate (or substrates). Due to this jigsaw puzzle-like match between an enzyme and its substrates enzymes are known for their specificity. The specificity results from the shape and the amino acid functional group's attraction to the substrate. There is a specifically matched enzyme for each substrate and, thus, for each chemical reaction.

The fact that active sites are so perfectly suited to provide specific environmental conditions also means that they are subject to local environmental influences. It is true that increasing the environmental temperature generally increases reaction rates, enzyme-catalyzed or otherwise. However, increasing or decreasing the temperature outside of an optimal range can affect chemical bonds within the active site in such a way that they are less well suited to bind substrates. High temperatures will eventually cause enzymes, like other biological molecules, to denature, a process that changes the substance's natural properties. Likewise, the local environment's pH can also affect enzyme function. Active site amino acid residues have their own acidic or basic properties that are optimal for catalysis. These residues are sensitive to changes in pH that can impair the way substrate molecules bind. Enzymes are suited to function best within a certain pH range, and, as with temperature, extreme environmental pH values (acidic or basic) can cause enzymes to denature.

You have learned that the activation energy required for many reactions includes the energy involved in manipulating or slightly contorting chemical bonds so that they can easily break and allow others to reform. Enzymes aid this process. The enzyme-substrate complex can lower the activation energy by contorting substrate molecules in such a way as to facilitate bond-breaking, helping to reach the transition state. Finally, enzymes can also lower activation energies by taking part in the chemical reaction itself. In these cases, it is important to remember that the enzyme will always return to its original state at the reaction's completion. One of enzymes' hallmark properties is that they remain ultimately unchanged by the reactions they catalyze. After an enzyme catalyzes a reaction, it releases its product(s).

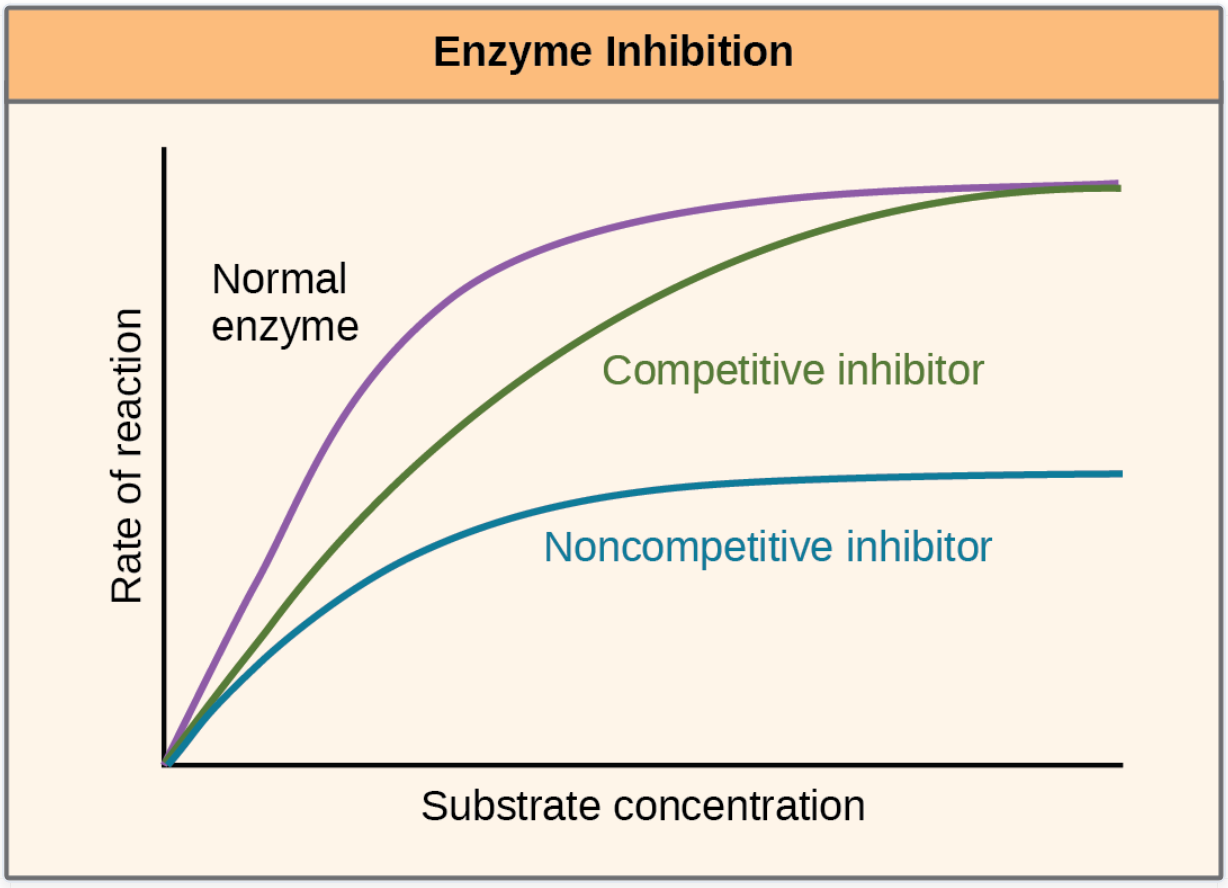


In this diagram, a substrate binds the active site of an enzyme and, in the process, both the shape of the enzyme and the shape of the substrate change. The substrate is converted to products that then leave the enzymes ...

Figure 6.16 According to the induced-fit model, both enzyme and substrate undergo dynamic conformational changes upon binding. The enzyme contorts the substrate into its transition state, thereby increasing the reaction's rate.

Molecular Regulation of Enzymes

Enzymes can be regulated in ways that either promote or reduce their activity. There are many different kinds of molecules that inhibit or promote enzyme function, and various mechanisms exist for doing so. For example, in some cases of enzyme inhibition, an inhibitor molecule is similar enough to a substrate that it can bind to the active site and simply block the substrate from binding. When this happens, the enzyme is inhibited through **competitive inhibition**, because an inhibitor molecule competes with the substrate for active site binding (Figure 6.17). On the other hand, in **noncompetitive inhibition**, an inhibitor molecule binds to the enzyme in a location other than the active site, called an **allosteric site**, which causes a conformational change in the shape of the enzyme, this reduces the enzyme activity as it no longer effectively catalyzes the conversion of the substrate to product.



This plot shows rate of reaction versus substrate concentration for an enzyme in the absence of inhibitor, and for enzyme in the presence of competitive and noncompetitive inhibitors. Both competitive and noncompetitive ...

Figure 6.17 Competitive and noncompetitive inhibition affect the reaction's rate differently. Competitive inhibitors affect the initial rate but do not affect the maximal rate; whereas, noncompetitive inhibitors affect the maximal rate.

EVERYDAY CONNECTION



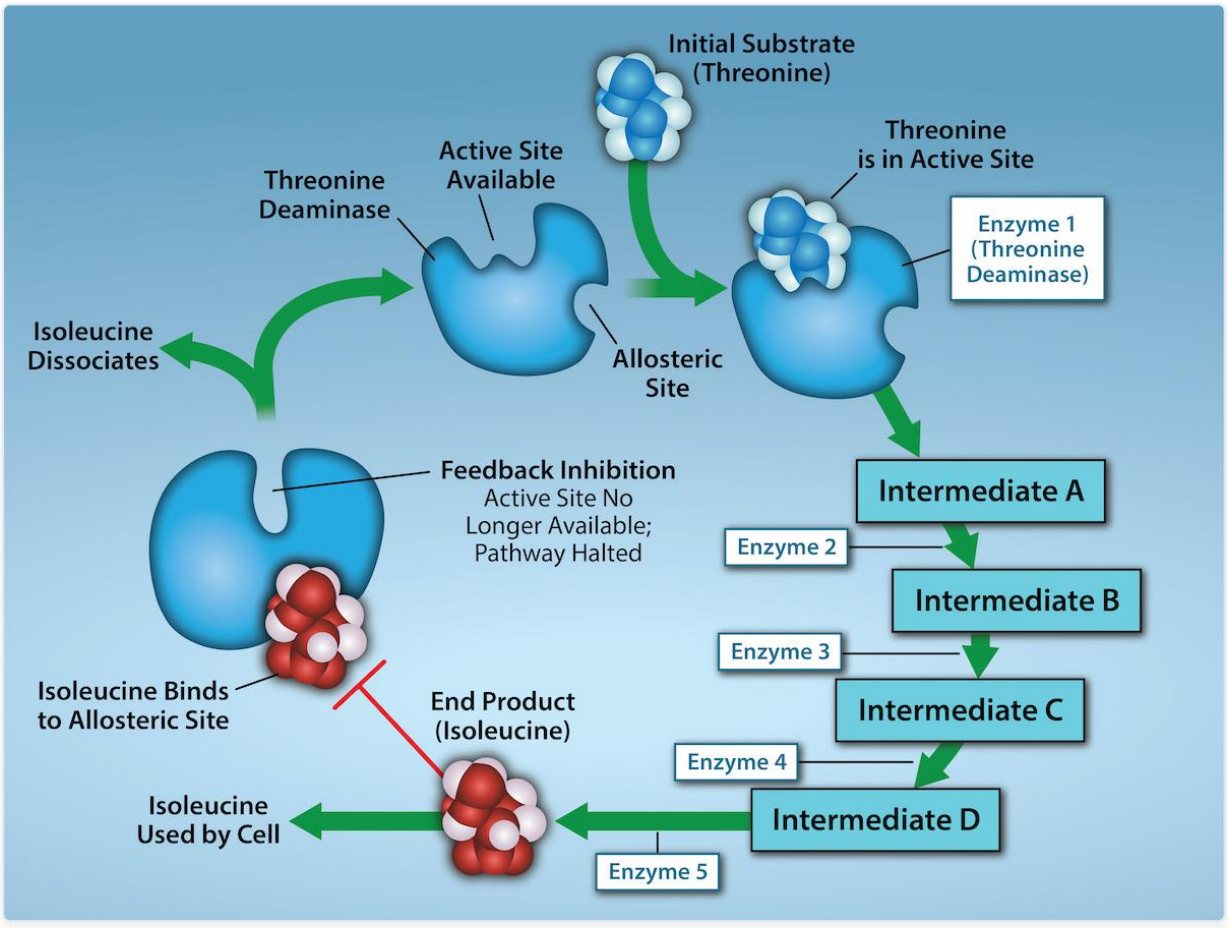
This photo shows several red capsule pills.

Figure 6.19 Have you ever wondered how pharmaceutical drugs are developed? (credit: Deborah Austin)

Consider statins for example—which is a class of drugs that reduces cholesterol levels. These compounds are essentially inhibitors of the enzyme HMG-CoA reductase. HMG-CoA reductase is the enzyme that synthesizes cholesterol from lipids in the body. By inhibiting this enzyme, the drug reduces cholesterol levels synthesized in the body. Similarly, acetaminophen, popularly marketed under the brand name Tylenol, is an inhibitor of the enzyme cyclooxygenase. While it is effective in providing relief from fever and inflammation (pain), scientists still do not completely understand its mechanism of action.

Feedback Inhibition in Metabolic Pathways

Molecules can regulate enzyme function in many ways. However, a major question remains: What are these molecules and from where do they come? Some are cofactors and coenzymes, ions, and organic molecules, as you have learned. What other molecules in the cell provide enzymatic regulation, such as competitive and noncompetitive inhibition? The answer is that a wide variety of molecules can perform these roles. The most relevant sources of enzyme regulatory molecules, with respect to cellular metabolism, are cellular metabolic reaction products themselves. In a most efficient and elegant way, cells have evolved to use their own reactions' products for feedback inhibition of enzyme activity. **Feedback inhibition** involves using a reaction product to regulate its own further production (Figure 6.21). The cell responds to the abundance of specific products by slowing down production during anabolic or catabolic reactions. Such reaction products may inhibit the enzymes that catalyzed their production through the mechanisms that we described above.



This diagram shows a metabolic pathway in which three enzymes convert a substrate, in three steps, into a final product. The image shows isoluecine binding to a allosteric site, which inhibits Threonine from binding to t...

Figure 6.21 Metabolic pathways are a series of reactions that multiple enzymes catalyze. Feedback inhibition, where the pathway's end product inhibits an upstream step, is an important regulatory mechanism in cells. Metabolic pathways are a series of reactions catalyzed by multiple enzymes (Intermediates A – D, Enzymes 1 – 5). Feedback inhibition occurs when the pathway's end product (here isoleucine) inhibits an upstream enzyme (indicated by red bar). In this example, isoleucine will bind to Threonine Deaminase (at the allosteric site) and prevent threonine from binding to this enzyme's active site, effectively blocking this metabolic pathway. When isoleucine levels decrease, Threonine will then bind to Threonine Deaminase's active site and the metabolic pathway will resume. This is an important regulatory mechanism in cells to inhibit the overproduction of a product. Credit: Rao, A., Ryan, K., Tag, A., Hawkins, A. and Fletcher S. Department of Biology, Texas A&M University.

Section 8.4: Instructor Lecture Slide Review (Chapter 8_students.pptx)

Official 36-slide lecture outline, core definitions, emphasis points, and energetic coupling rules.

Slide 1–6: Energy Currency & ATP Hydrolysis

A cell is a chemical factory. ATP (Adenosine Triphosphate) is the primary energy currency composed of adenine nitrogenous base, ribose sugar, and three phosphate groups. Hydrolysis of the phosphoanhydride bond between the 2nd and 3rd phosphate releases free energy (ATP + H_2O → ADP + P_i + Energy out). Condensation/phosphorylation requires energy input (Energy in + ADP + P_i → ATP + H_2O).

Slide 7–12: Cellular Metabolism & Carbon Sources

Metabolism includes all biochemical reactions maintaining cellular life. *Autotrophs* ("self-feeders") synthesize organic molecules from inorganic carbon; *photoautotrophs* convert sunlight into triose precursor G3P. *Heterotrophs* ingest and hydrolyze organic polymers synthesized by autotrophs. Pathways alter starting reactants through stepwise intermediates into products.

Slide 13–20: Gibbs Free Energy (ΔG) & Pathway Energetics

Gibbs Free Energy (G) is usable energy available to do cellular work. Reactions with -ΔG are *exergonic* (spontaneous, downhill, release energy, products have less potential energy). Reactions with +ΔG are *endergonic* (non-spontaneous, uphill, require energy payment of ATP, products have more potential energy). Anabolic pathways are endergonic (+ΔG); catabolic pathways are exergonic (-ΔG).

Slide 21–27: Activation Energy (E_A) & Coupled Reactions

What prevents exergonic reactions from spiraling out of control? The Energy of Activation (E_A) barrier! Reaching the transition state requires stretching and contorting chemical bonds. Enzymes lower E_A without altering ΔG. Practice problem from slides: Reaction A → B (ΔG = +321) coupled with S → T (ΔG = -125) CANNOT occur spontaneously because net ΔG = +196 > 0.

Slide 28–36: Enzyme Mechanics, Inhibitors & Feedback Control

Enzymes bind substrates at their active site via induced fit. Enzymes end in "-ase". Environmental extremes (high temperature, acidic/basic pH, salts) denature tertiary protein structure. Cofactors (inorganic metal ions, vitamin-derived coenzymes) aid catalysis. *Competitive inhibitors* mimic substrate shape and block the active site (reversible with high [S]). *Noncompetitive inhibitors* bind allosteric sites, changing enzyme conformation. *Feedback inhibition* occurs when a pathway end product acts as an allosteric inhibitor on an upstream enzyme to stop overproduction.

Section 8.5: Official Course Practice Questions & Solutions

Original 24 practice examination problems and 4 energetic coupling challenge sets with answer keys.

Part A: Chapter 8 Practice Problems (24 Questions)

1. Only about 40% of the chemical energy stored in a glucose molecule can be accounted for in ATP molecules that are generated during cell respiration. The rest turns to heat energy. What is this an example of?

A) First Law of Thermodynamics • **B) Second Law of Thermodynamics (Correct)** • C) Third Law • D) None of the above

2. Which of the following describes an anabolic pathway?

A) All reactions for life • B) Releases ATP & breaks polymers • **C) Reactions that require ATP & build polymers from monomers (Correct)** • D) None of the above

3. Which of the following describes a catabolic pathway?

A) All reactions for life • **B) Reactions that release ATP & break down polymers into monomers (Correct)** • C) Requires ATP & builds polymers • D) None of the above

4. If a reaction is spontaneous (exergonic), is the value of ΔG positive or negative?

A) Positive • **B) Negative (Correct)**

5. An endergonic reaction has which of the following properties?

A) +ΔG, spontaneous • **B) +ΔG and the reaction is not spontaneous (Correct)** • C) -ΔG, spontaneous • D) -ΔG, not spontaneous

6. Which of the following is one way that ADP is converted to ATP?

A) A free inorganic phosphate group is added to ADP (Correct) • B) Hydrolysis • C) Not possible • D) Both A & B

7. A covalent bond between two atoms represents what kind of energy?

A) Kinetic • **B) Potential (Chemical) (Correct)** • C) Mechanical • D) Solar

8. What role do enzymes play in a chemical reaction?

A) Slow down • B) Create reactions • C) Destroy reactions • **D) Speed up a chemical reaction (Correct)**

9. What is a substrate?

A) What enzymes react on, the reactant in enzymatic reactions (Correct) • B) Product • C) Enzyme itself • D) Inhibitor

10. How do enzymes speed up chemical reactions?

A) Increase activation energy • **B) Lower activation energy (Correct)** • C) Unknown • D) Decreases total energy change

11. Which of the following could be a cause of a denatured enzyme?

A) Competitive inhibitor • B) Non-competitive inhibitor • **C) A spike in pH (Correct)** • D) Inability of substrate to fit

12. Which is the most likely result of a non-competitive inhibitor?

A) The reaction slows down (Correct) • B) Reaction speeds up • C) Active site is blocked directly • D) Enzyme denatures

13. What is a competitive inhibitor?

A) An inhibitor that moves into the active site of the enzyme (Correct) • B) Moves into non-active site • C) Denatures enzyme • D) Destroys substrate

14. Which of the following is not true concerning ATP?

A) Formed by attaching phosphate with high energy bond • B) Only outer bond hydrolyzed • **C) It is a good long-term energy storage molecule (Correct - glycogen/starch are long-term)** • D) Becomes ADP when dephosphorylated

15. Which of the following is not true of enzymes?

A) Use 3D shape of active site • B) Lower activation energy • **C) Enzymes make ΔG more negative (Correct - ΔG is unchanged)** • D) Function best at specific temperature and pH

16. A roller coaster is sitting at the top of a hill, about to drop off. Which of the following best describes the energy present?

A) Potential energy (Correct) • B) Chemical energy • C) Kinetic energy • D) Heat energy

17. In an endergonic reaction, do the products or reactants have more energy?

A) Reactants • **B) Products (Correct)**

18. Feedback inhibition is an efficient way to control a metabolic pathway because:

A) First enzyme inhibited by own product • B) Last enzyme inhibited by own product • **C) The first enzyme in the pathway is inhibited by the end-product of the pathway (Correct)** • D) Last enzyme inhibited by end-product

19. Where is the energy stored in a molecule of ATP?

A) Between nitrogen and carbon • B) In ribose C-C bonds • C) In P=O double bond • **D) In the phosphoanhydride bond connecting the second and third phosphate groups (Correct)**

20. Coenzymes and cofactors assist protein function by:

A) Stabilizing 3D shape only • **B) Assisting with the binding between the substrate and the active site (Correct)** • C) Blocking active site • D) Denaturing enzyme

21. Explain why amylase works in the mouth but not in the stomach:

Answer: Amylase has an optimum pH near neutral (≈ pH 7), matching saliva. The stomach is highly acidic (≈ pH 2), which alters active site ionization and denatures the amylase enzyme.

22. For reaction A → B (ΔG = -200), which reaction could be coupled?

A) C → D (ΔG = +220) • B) E → F (ΔG = -125) • **C) G → H (ΔG = +180) (Correct: Net = -200 + 180 = -20 < 0)** • D) K → L (ΔG = -250)

23. For reaction Y → Z (ΔG = +225), which reaction could be coupled?

A) C → D (ΔG = +220) • B) E → F (ΔG = -125) • C) G → H (ΔG = +180) • **D) K → L (ΔG = -250) (Correct: Net = +225 + (-250) = -25 < 0)**

Part B: Reaction Coupling Practice Problems (4 Scenarios)

Scenario 1: Reaction B → C (ΔG = -100)

- **Thermodynamic Type:** Exergonic (ΔG < 0)
- **Metabolic Type:** Catabolic (releases energy)
- **Coupled Option: A reaction with ΔG = +90** (Net ΔG = -100 + 90 = -10 < 0; spontaneous).

Scenario 2: Reaction G → H (ΔG = +50)

- **Thermodynamic Type:** Endergonic (ΔG > 0)
- **Metabolic Type:** Anabolic (requires energy payment)
- **Coupled Option: A reaction with ΔG = -100** (Net ΔG = +50 + (-100) = -50 < 0; spontaneous).

Scenario 3: Reaction L → M (ΔG = +500)

- **Thermodynamic Type:** Endergonic (ΔG > 0)
- **Metabolic Type:** Anabolic (requires energy payment)
- **Coupled Option: A reaction with ΔG = -510** (Net ΔG = +500 + (-510) = -10 < 0; spontaneous).

Scenario 4: Reaction Y → Z (ΔG = -35)

- **Thermodynamic Type:** Exergonic (ΔG < 0)
- **Metabolic Type:** Catabolic (releases energy)
- **Coupled Option: A reaction with ΔG = +25** (Net ΔG = -35 + 25 = -10 < 0; spontaneous).

Appendix: Course Active Learning Practice Sets & Solutions

Original in-class problem sets, thermodynamic coupling calculations, and structural inhibition models from Active Learning #3 (Slides 2–3).

Problem 1: Reaction Coupling (Thermodynamics & Delta G) • Active Learning #3, Slide 2

Determine the thermodynamic feasibility of energetic coupling. Energetic coupling requires the net reaction to be exergonic ($\Delta G_{net} = \Delta G_1 + \Delta G_2 < 0$):

Reaction Scenario	ΔG Value	Thermodynamic Type	Valid Coupled Reaction & Rationale
Reaction A → B	ΔG = -200 kcal/mol	**Exergonic** (Releases energy)	**I → J (ΔG = +175 kcal/mol)** ΔG_{net} = -200 + 175 = -25 kcal/mol (< 0, Spontaneous). The energy released by A → B exceeds that required by I → J.
Reaction X → Y	ΔG = +75 kcal/mol	**Endergonic** (Requires energy)	**C → D (ΔG = -125 kcal/mol)** ΔG_{net} = +75 + (-125) = -50 kcal/mol (< 0, Spontaneous). An exergonic driver releasing ≥ 75 kcal/mol is required.

Problem 2: Structural Enzyme Inhibition • Active Learning #3, Slide 3

Biological Context: Acetylcholinesterase degrades the neurotransmitter acetylcholine to allow muscle relaxation. Analyze the mechanism of two distinct inhibitors:

Inhibitor	Structural Binding Site	Inhibition Class	Mechanism & Functional Consequence
Sarin	Allosteric Site (outside active site)	**Noncompetitive (Allosteric)**	Binds to a distinct regulatory pocket, inducing a conformational change that distorts the active site notch. Substrate cannot bind regardless of [S].
Mullerian	Active Site (V-shaped catalytic pocket)	**Competitive**	Directly mimics acetylcholine's geometric shape; physically blocks substrate access. Can be overcome by high substrate concentrations.